Maintenance, Storage, and Export of Pacific Masters Pool Records
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Overview: Currently the Pacific Masters records are stored in a HyTek database on the Pacific Masters Administrator's computer. Records are updated as they are discovered, from various sources. An export from the database is uploaded to the Pacific Masters website where everyone can see them. This export is not satisfactory as it truncates names to 17 characters and contains many inconsistencies. The purpose of this document is to specify the future method for storing and maintaining the records, both individual and relay.

Storage: The Pacific Masters pool records will be stored in the Pacific Masters database pacmasters_data on the current hosting server for the Pacific Masters website. Current records, historical records, and nullified records will be kept. The database has been designed to force consistency of names for individual record holders by keeping a separate table of record holder names. The database table layout is available in this document: https://docs.google.com/spreadsheets/d/1Eg4lJqzjCvR7rGo8G-jsCLqb0rtnOcy6G_4ggbVhIlA/edit#gid=0

***NOTE: The records database on the production server should now be considered the official repository for records. The tables have been normalized and name spellings corrected and made consistent. Testing can be done on the dev server.

Inputs: There will be two ways of updating the PacM records database. The first will be automated through the DrPacM process, as an SQL insert. The second way to update the records will be through a web form, accessible only through logins on the Pacific Masters website, so that authorized people will be able to manually input records. Assuming that the DrPacM process captures all the new individual records, the web form will be used mostly for new relay records. Both input methods will use a PHP function to do the insert into the database. This will be described below.

Manual input for individual records

A web form will be created to allow manual input for records. The fields will be:

Course
Gender
Age Group
Distance
Stroke
Swimmer First Name
Swimmer MI (if any)
Swimmer Last Name
Swimmer ID (last five characters)

Swimmer Club (if known)
Date of swim
Time (duration) of swim
Splash ID (if known)

Course, Gender, Age Group, Distance, and Stroke will be presented as drop-down boxes listing the valid choices. The swimmer information (name, ID, and club) can be compared with the membership table, if the membership list is stored and kept up-to-date in the web server database. This way mismatches can be flagged, thus maintaining a high level of data integrity. If the membership list is not kept in the database, then the probability of errors is higher.

Assuming the membership table is available, then if there is a data mismatch, the information that doesn't match will be displayed, and the user will have the choice of keeping the original data, or entering new information.

Manual input for relay records

The manual entry of relay records will be the same as for individual records with some differences. The "stroke" will be limited to Free or Medley, the age groups will be standard relay age groups, and there will be four names to enter. The swimmer ID is not necessary for relays, but the club name is.

PHP function for adding records to the database

Both the DrPacM and the manual entry process will use this function to add records to the database. The function does some basic error checking, and returns the result as an array of messages as strings. Messages with "Fatal:" at the beginning will prevent any modifications to the database. Message without "Fatal:" are warnings, but will not cause processing to stop. The function is located in ~/Library/pacmfncn.php and requires ~/Library/pacminc.php.

The function call is as follows:
add_ind_record($course, $gender, $age_group, $distance, $stroke, $name, $sid, $club, $rdate, $rtime, $splashid, $userid)

$course - one of LCM, SCM, SCY (case insensitive, required)
$gender - one of M or F (case insensitive, required)
$age_group - one of DD-DD or DDD-DDD, standard age groups, required
$distance - 50, 100, 200, 400, 500, 800, 1000, 1500, 1650 depending on course, required
$stroke - must be one of Free, Back, Breast, Fly, I.M. exactly as spelled here, required
$name - full name of record holder, required
$sid - last 5 characters of USMS swimmer ID, case insensitive, must be alphanumeric and exactly 5 characters long

$club - club abbreviation, case insensitive, can be empty string
$rdate - date of record swim as YYYY-MM-DD, all digits
$rtime - time (duration) of record swim as hh:mm:ss.tt, must be 11 characters, only digits and ":" and "."
$splashid - there are no restrictions on this yet
$userid - if coming from DrPacM, this must be "DrPacM", otherwise it is the login name of a person doing manual adds through the website

After a successful addition to the records table, a transaction will be entered into the log table. The information captured will be userid, the unique record id generated by the insertion into the records table, the type of transaction (in this case, an addition), the reason for the transaction (in this case, addition from web form or DrPacM), the record type (this function is for individual records), and the transaction timestamp.

In the case of the web form, the information initially submitted with the form will be checked, and then displayed back to the user for confirmation. If there are fatal errors, these will be displayed (along with any warning messages) and the user must correct the problem by going back to the initial form. If there are no errors, the information is displayed again with a "Confirm" button. Clicking on the confirm button will enter the information into the database.

The web form will be accessed from the Database (Admin) page on the website, which is under the submenu "Admin Tasks", and is only visible to, and can only be accessed by, authorized users.

Output: There will be three methods of exporting the records:

1. Export to Meet Manager
2. Export for DrPacM
3. Export for AGSOTY

Export to Meet Manager for Individual Records is as follows. The records are exported in files named IPAC.lcm, IPAC.scm, IPAC.scy, with the suffix identifying the corresponding course. The format of these files is one record per line:

GGGGG,AG-AG,DDDD U. SSSS,First Last ,MM/DD/YY,MM:SS.HH

- GGGGG is WOMEN or M E N (5 characters)
- AG-AG is the age group (5 characters)
- DDDD is the distance (5 characters, left justified, padded with spaces)
- U. is the units, either YD. or M. (3 characters, left justified, padded with spaces)
- SSSS is the stroke, one of FREE, BACK, BRST, FLY, I.M. (4 characters, right justified)
- First Last is the full name (17 characters only, left justified, padded with spaces)
- MM/DD/YY is the date of the record swim (8 characters, leading 0's included)

- MM:SS.HH is the duration of the swim to hundredths of a second, with leading 0's omitted (8 characters, right justified)

Note: if any records are ties, the names will need to be scrunched together in the 17 character field.

Export to Meet Manager for Relay Records is as follows. The records are exported in files named RPAC.lcm, RPAC.scm, RPAC.scy, with the suffix identifying the corresponding course. The format of these files is one record per line:

GGGGG,DDDD U. SSSSSS,AG-AG,CLUB,N1,N2,N3,N4,MM/DD/YY,MM:SS.HH

- GGGGG is one of WOMEN, M E N, or MIXED (5 characters)
- AG-AG is the age group, 5 characters if it is 72-99, otherwise 7 characters
- DDD is the distance (5 characters, left justified, padded with spaces)
- U. is the units, either YD. or M. (3 characters, left justified, padded with spaces)
- SSSSSS is either FREE or MEDLEY (6 characters, right justified, padded with spaces)
- CLUB is 15 characters, all uppercase, left justified
- N1 is the name of the first swimmer, 17 characters, left justified
- N2, N3 are the second and third swimmers, 17 characters, left justified, but with a leading space
- N4 is the fourth swimmer, is also left justified with a leading space, but is 27 characters
- MM/DD/YY is the date of the record swim (8 characters, leading 0's included)
- MM:SS.HH is the duration of the swim to hundredths of a second, with leading 0's omitted (8 characters, right justified)

Note: the names of the relay swimmers are in many formats: "First Last", "F Last", "Last", "F M Last", and sometimes just a single initial.
Note: the leading spaces on the second, third, and fourth name make the print out of the meet results look better.

Individual records extraction for AGSOTY and DrPacM

A PHP function is available for extracting individual records from the database. This function is called "**ind_records_extract**" and takes one argument, the course (one of LCM, SCM, SCY). This function will extract all the records, regardless of status, and will return an array of associative arrays with each of the associative arrays having the following values:

- gender: M or F
- age_group: standard age groups in the form DD-DD (or DDD-DDD for age groups over 100)
- distance: standard numerical event distances
- stroke: full name of stroke (Freestyle, Butterfly, Backstroke, Breaststroke, Individual Medley)
- date: date of swim in the form YYYY-MM-DD

- duration: the record time in the form hh:mm:ss.tt
- ftime: the status of the record, as follows:
    - 0 - historical
    - 1 - current fastest
    - 2 - unverified (may or may not become fastest)
    - 3 - nullified
    - 4 - unverified historical (an unverified record that was broken before it was discovered, or that was not verified before a faster swim was added)
    - 5 - fails standards (a record that would be the current fastest record if it had passed standards)
- name: full name of swimmer
- swimmer_id: last 5 characters, all uppercase, of the USMS swimmer id at time of swim, if available
- splash_id: a unique identification of the splash that set the record, as the last part of the URL for the splash on the USMS website, if available
- course: this is the input variable used to select the records

The extracted records are ordered as follows:

- Gender (M, then F)
- Stroke (in the order Freestyle, Backstroke, Breaststroke, Butterfly, Individual Medley)
- Distance (shortest to longest)
- Age Group (youngest to oldest)

Any failure will result in the return of an array consisting of a single string with an error message.

Display:
Currently the website displays one of six tables - current relay and current individual records for each of the three courses. The following enhancements will be applied:

1. The ability to view historical records
2. Converting the time field to a link to the splash information page on the USMS website, if the splash ID is available
3. An asterisk to indicate that a record has not been verified

Verification of records:
New records input by the DrPacM method will be marked as unverified. An email notification will be sent to the responsible parties alerting them to the new record. Once the record is verified, it will need to be marked as such in the database. This will be accomplished through a simple(!) web form that will display all unverified records and provide a checkbox for each record which the user can check to make the change. These changes will be logged in the transaction log.

Nullification of records:
A method will be designed to nullify recent record entries, regardless of status. This will be a form accessible from the Database (Admin) page. All records less than two(?) years old will be

displayed for a particular record type and course. Any record selected for nullification will have ftime set to 3, an entry will be made in the records transaction log, and the most recent verified historical time will become the fastest time.

Tracking metadata:
For each modification to the records tables, a transaction will be logged in a separate table. This table will log the following:

- Who: either Drupal username or DrPacM
- Date of transaction
- Transaction type: new entry, verification of automatic entry, nullification of record, change of status from fastest to historical
- Reason for transaction: needed only for nullification and manual adds(?)
- Record type (relay or individual), course, and record id